SCIENCE / BC GRADE 9 / 10

Balance chemical equations

Track atoms as substances react and rearrange.

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01 Atoms are conserved

In an ordinary chemical reaction, atoms rearrange into new combinations. The balanced equation contains the same number of atoms of each element on each side. Mass is conserved in a closed system.

02 Coefficients and subscripts

A coefficient multiplies the entire formula after it. A subscript belongs to the substance's formula. Change coefficients to balance; changing a subscript changes the substance.

03 A reliable method

Write correct formulas first. Count atoms by element, adjust coefficients, and recount. Reduce coefficients to the smallest whole-number ratio. State symbols describe physical state but do not change atom counts.

Worked example

Balance $H_2 + O_2 \rightarrow H_2O$

Oxygen appears twice on the left and once on the right. Put 2 before H_2O; now the right has four H atoms. Put 2 before H_2. Final: $2H_2 + O_2 \rightarrow 2H_2O$. Both sides contain four H atoms and two O atoms.

Watch out: Do not turn H_2O into H_2O_2 to balance oxygen. That would describe a different substance.

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Try it. Then check it.

Work through these questions before looking at the answers.

1. Balance $Mg + O_2 \rightarrow MgO$.

2. How many H and O atoms are represented by $3H_2O$?

3. A sealed reaction vessel has mass 120.0 g before a reaction. Predict its mass afterward if nothing enters or leaves.

Answer key

1. $2Mg + O_2 \rightarrow 2MgO$. Both sides contain two Mg atoms and two O atoms.

2. Six H atoms and three O atoms. The coefficient multiplies both elements.

3. 120.0 g, within measurement uncertainty, because mass is conserved.

One-minute recap

Close the notes. Explain one key idea in your own words, then retry the question you found most difficult.

About this guide

Original Study Bridge topic review prepared with AI assistance. Suggested course fit, not complete or official course coverage. Compare with your teacher's materials. Examples and practice questions are newly written.

Official course context: Science 9-10 · chemistry foundations

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